

Hima — Support Bot

AI ticket-deflection feature — **how it works**, configuration, and operations

Innovfix Pvt. Ltd. · Hima App · v1 · 2026-07-17 · Backend: Laravel (demo_hima / hima-admin-panel) · App: Android (hima_app)

What it is, in one line. A guided in-app chat that sits **in front of** the existing ticket system: it asks the user 3–4 questions, reads **their own live account data**, answers, and only creates a support ticket if the user says the answer did not help. The goal is to stop the repeat-contact tickets (the average user raised ~2 per month) at the source — without ever making a false statement about someone's money or account.

1. The user flow

Step	Screen	Notes
0 · Language	"Choose your language" (prompt in English, all 11 options)	Applies to this conversation only — does not change the profile language / matching.
1 · Category	5 categories, volume-ordered	Calls · Earnings/Withdrawal · Coins/Recharge · Account/Profile · Something else.
2 · Sub-category	The category's sub-options + "My issue isn't listed"	Chips are server-driven — the app renders labels and posts keys; it knows nothing about the taxonomy.
3 · One detail	A sub-category-specific question (optional, with Skip where safe)	e.g. "double charge" asks for a date/time; the answer genuinely needs it.
4 · Answer	Typing → the AI's answer	Only this step is slow. The bot has already read the user's data before replying.
5 · Did this solve it?	[Yes, solved] · [No, still need help]	Yes → close, no ticket . No → one more attempt, then a human.

Yes vs No — the whole point. "Yes" closes the chat and **creates no ticket row** (a deflected issue must not appear in tickets, or deflection is unmeasurable). "No" buys **one** second, different answer; a second "No" creates a ticket for a human, pre-tagged and with the bot's diagnostics attached. There is no third attempt.

2. How it stays accurate (why it can be trusted with money)

- **Reads live data via fixed, read-only tools** — never free-form SQL. Eight tools: account summary, recent calls, call detail, recharge history, coin ledger, withdrawal history, earnings summary, block status.
- **The user is structural, not a parameter.** The logged-in user id is fixed on the server; it appears in **no** tool the model can call, so the model cannot express "look at another user's data" (closes IDOR by construction).
- **Grounding guard.** Every number in a reply (₹ amounts, coins, order ids, times) must be a value a tool actually returned. Any invented number → the answer is discarded and the issue is escalated.
- **Skip-AI on money.** For sensitive/redirect money categories the model is never called at all — a human takes it directly, so money cannot be hallucinated about.
- **Recharge matching is conservative.** A payment is stated as "credited" only when a credit of the exact expected size sits in its time window and belongs to no other payment; anything ambiguous is marked for human review, never asserted.
- **Fail closed.** If a tool cannot read the user's data (DB/replica down), the bot escalates to a human rather than answering around the gap. On prod, reads use a replica; if the replica lags, it falls back to the primary so it never states stale data.
- **Never promises** a refund, credit or timeline; **never deletes** an account — a delete request is treated as a symptom and probed.

3. Configuration & kill-switches (ops)

Setting	Where	Effect
support_bot_enabled	gateway_config	Master kill-switch. When set to <code>0</code> , the app shows the old raise-ticket form and in-flight sessions escalate to a human instead of calling the model. Fails closed (unreadable = off).
support_bot_model	gateway_config	The model id (default Claude Sonnet). Deliberately separate from the ticket-answering model.
known_issue_banner	gateway_config	When set, shown first during a platform incident — deflects the duplicate flood before the menu.
support_hours_start / _end	gateway_config	Agent shift (default 4 PM–12 AM IST) for the out-of-hours "when we'll look" ETA.
SUPPORT_BOT_READ_CONNECTION	.env	Read replica for tool queries on prod (empty on demo). Lag-guarded.
SUPPORT_BOT_SESSION_CEILING	.env	Hard per-session deadline (25s) — under PHP's execution limit; the app waits longer so it always gets the definitive answer.
SUPPORT_BOT_TOKEN_BUDGET	.env	Hard per-session cost ceiling; exceeding it escalates rather than spending more.

Answers are ops-editable without a deploy. The per-sub-category playbooks live in `ai_reference_faqs` (already per-language, already an admin UI). The Yes/No taps are a free labelled dataset — a sub-category below ~40% "Yes" is a broken playbook or a real product bug.

4. Endpoints (all `POST`, no `user_id` ever accepted from the client)

Endpoint	Purpose
support_bot_start	Opens/*resumes* a session; returns language chips + any known-issue banner. Checks for a pending escalated ticket up front.
support_bot_reply	Walks the tree, then answers. One request per session at a time (locked).
support_bot_feedback	"Did this solve it?" — the only thing that resolves or escalates. Runs the second attempt.
support_bot_session	Rebuilds a session reopened after the app was killed (transcript + current step).
support_bot_attach	Optional file on the ticket the bot raised (image/video/audio, ≤5MB). Serialised per ticket.

5. Rollout & what to watch

- Demo-first, always.** Everything is verified on `demohima.himaapp.in` before prod. Prod goes on an explicit go, staged by sub-category allowlist and user %.
- Fallback is safe.** With the kill-switch off, or on any start failure, the user gets the old form — the support channel is never removed behind the LLM.
- Weekly metrics:** per-sub-category Yes-rate · grounding-guard trips (should be ~0) · p50/p95 latency · tokens/session · **48h re-contact rate** · and the north star: **tickets per user per month**.

Design guardrail — "calls not coming" (the #1 driver) is a real product bug the bot cannot fix. The bot reads the data and escalates; it does **not** troubleshoot ("check your network"), which reads as being fobbed off by a robot. Its deflection target there is near-zero **by design** — track it as a product-bug signal, not a deflection KPI.

6. QA test cases (run before release)

Test as a **male** and a **female** account, on WiFi and mobile data. The **MONEY** block is MUST-PASS — verify every figure against the DB, never the reply alone.

✓	Test
☐	Flow: language (all options, prompt in English) → category (5) → sub-category (+ "My issue isn't listed") → optional detail → answer → "Did this solve it?" Yes/No.
☐	Language stays put for the whole chat (no silent switch); reply mirrors the script the user typed (Tanglish in → Tanglish out).
☐	Sub-category follow-up is specific (double-recharge asks a date/time), not the same generic prompt for all.
☐	Deflection: Yes → closes, optional rating, NO ticket row (confirm in DB).
☐	Escalation: No → one <i>different</i> second attempt → No again → ticket to a human with diagnostics.
☐	Delete-account request is probed, never actioned; the bot never offers deletion.
☐	Resume mid-flow: force-kill the app on any step → reopen → rebuilds the conversation on the same step (no re-answering).
☐	Resume after answer: shows THAT answer with Yes/No (never the rejected first answer); resolved/escalated sessions reopen read-only.
☐	Attachment: optional, ≤5MB, image/video/audio; too-large/wrong-type rejected with a clear message; a rejected upload does not look accepted; a second pick while uploading is blocked; rotating the screen mid-upload does not break it.
☐	Slow network: the app waits (~30s) for the definitive answer/escalation — first attempt AND the second — never fails early to the form.
☐	Kill-switch: <code>support_bot_enabled=0</code> → old form; flip OFF mid-session → next step escalates to a human, no hang.
☐	Pending ticket: a user with an open escalated ticket is told at the start ; no duplicate is created (even from two devices at once).
☐	Zoho launcher hidden on the bot screen for female users and restored on exit; never appears for male users.

MONEY — MUST-PASS (verify against the DB).

- **Recharge:** for a real ₹1–₹49 recharge (including one completed >30 min after starting), "credited / not credited" matches the DB; two close purchases are never mis-attributed — ambiguous cases say a human will check, never a false "you were credited".
- **No invented numbers** — every ₹, coin, order id or time came from the data; no unsupported refund/timeline.
- **Earnings "connected"** excludes `00:00:00` (matches Recent Calls).
- **IDOR:** a session/ticket only ever reflects the logged-in user's own data.